A FINITE MORPHISM WITH A SINGLETON FIBRE IS LOCALLY DETERMINED NEAR THE POINT, WITH FINITE LOCAL RINGS AND MATCHING STALKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13551-13568 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13570-13591, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Workhorse lemma used by the chapter's etale-local structure arguments; its parenthetical (some details omitted) covers only routine localization properties while every claim is named.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Basic properties of finite morphisms (closed, affine), the unique-prime-over localization lemma (Algebra, lemma-unique-prime-over-localize-below), and elementary localization identities for modules.

Source reference. Original lemma 00EA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13591–13591. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite morphism. Let $x \in X$ with $y = \pi(x)$ such that $\pi^{-1}(\{y\}) = \{x\}$. Then*

1. *For every neighbourhood $U \subset X$ of x in X, there exists a neighbourhood $V \subset Y$ of y such that $\pi^{-1}(V) \subset U$.*
2. *The ring map $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ is finite.*
3. *If π is of finite presentation, then $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ is of finite presentation.*
4. *For any quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ we have $\mathcal{F}_x = \pi_*\mathcal{F}_y$ as $\mathcal{O}_{Y,y}$-modules.*

Proof. The first assertion is purely topological; use that π is a continuous and closed map such that $\pi^{-1}(\{y\}) = \{x\}$. To prove the second and third parts we may assume $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$. Then $A \to B$ is a finite ring map and y corresponds to a prime $\mathfrak{p}$ of A such that there exists a unique prime $\mathfrak{q}$ of B lying over $\mathfrak{p}$. Then $B_\mathfrak{q} = B_\mathfrak{p}$, see Algebra, Lemma 00EA. In other words, the map $A_\mathfrak{p} \to B_\mathfrak{q}$ is equal to the map $A_\mathfrak{p} \to B_\mathfrak{p}$ you get from localizing $A \to B$ at $\mathfrak{p}$. Thus (2) and (3) follow from simple properties of localization (some details omitted). For

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

the final statement, suppose that $\mathcal{F} = \widetilde{M}$ for some B-module M. Then $\mathcal{F} = M_{\mathfrak{q}}$ and $\pi_*\mathcal{F}_y = M_{\mathfrak{p}}$. By the above these localizations agree. Alternatively you can use part (1) and the definition of stalks to see that $\mathcal{F}_x = \pi_*\mathcal{F}_y$ directly. $\square$